LAB ACTIVITY 3 – PART 1: THE DATA FOUNDATION

Topic: 3.1.1 JavaScript Variables, Data Types, and Operators

Duration: 45–60 Minutes

Student Name	**MOHAMAD ADAM HAIDIL BIN MOHD ADNAN**
Registration No	**10DIT25F1053**
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1. Learning Outcomes

By the end of this session, you should be able to:

1. **Declare** variables using let and const appropriately.
2. **Identify** different JavaScript data types (String, Number, Boolean, Array).
3. **Perform** calculations using Arithmetic and Assignment operators.
4. **Display** data using console.log() and alert().

2. Part A: Theory (The Building Blocks)

Concept	Description	Example
Variables	Containers for storing data values.	let price = 10;
Strings	Textual data wrapped in quotes.	"Hello World"
Numbers	Numerical data (integers or decimals).	25 or 3.14
Booleans	Logical values: True or False.	isLoggedIn = true;
Operators	Symbols used to perform operations.	+, -, *, /, %

3. Part B: Practical Step-by-Step

Step 1: Variable Declaration & Setup

Goal: Set up the file structure and define basic variables.

1. Create a folder named **RegNo_Lab3_Part1**.
2. Create an index.html file.
3. Inside the <body>, insert the Header and the first Lab Card:

HTML

```
<header>
    <h1>Lab 3.1.1: JavaScript Fundamentals</h1>
    <p>Variables, Data Types, and Operators</p>
</header>

<section class="lab-card">
    <h2>Steps 1 - 3: Logic & Console</h2>
    <div class="instruction">
        Open the <strong>Browser Console</strong> (F12 > Console) to see
the output for variables and <code>typeof</code> checks.
    </div>
    <button onclick="runConsoleTasks()">Run Console Tasks</button>
    <p class="note">*This logs Course Name, Student Count, and Data
Types.</p>
</section>
```

4. Inside the <script> tag, define the first function covering **Steps 1, 2, and 3**:

JavaScript

```
/* ==========================================
   STEPS 1, 2 & 3: Variables & Operators
   ========================================== */
function runConsoleTasks() {
    // Variable Declaration
    const courseName = "Web Design Technology";
    let studentCount = 30;
    let isLabFinished = false;

    console.log("--- Step 1: Variables ---");
    console.log("Course:", courseName);
    console.log("Initial Students:", studentCount);

    // Arithmetic
    let x = 10;
    let y = 3;
    console.log("--- Step 2: Arithmetic ---");
    console.log("Addition (10+3):", x + y);
    console.log("Remainder (10%3):", x % y);

    // Type Identification
    console.log("--- Step 3: Data Types ---");
    console.log("Type of courseName:", typeof courseName);
    console.log("Type of studentCount:", typeof studentCount);
    console.log("Type of isLabFinished:", typeof isLabFinished);

    alert("Check your browser console (F12) to see the results!");
}
```

5. Capture a screenshot of the output from the previous steps. **Observe** the results and **explain** your understanding of the process.

Screenshot of the output
Explanation

Step 4: Simple UI Interaction (Prompt & Alert)

Goal: Interact with the user using built-in browser dialog boxes.

1. Add this new section to your HTML <body>:

HTML

```
<section class="lab-card">
    <h2>Step 4: Simple UI Output</h2>
    <div class="instruction">
        Test the <code>prompt()</code>, <code>alert()</code>, and
<code>confirm()</code> functions.
    </div>
    <button onclick="runInteraction()">Start Interaction</button>
</section>
```

2. Add the function to your <script> tag (below runConsoleTasks):

JavaScript

```
/* ==========================================
   STEP 4: Interaction
   ========================================== */
function runInteraction() {
    let username = prompt("What is your name?");
    if (username) {
        alert("Welcome to JS Lab, " + username + "!");

        let proceed = confirm("Do you want to start the lab?");
        console.log("User wants to proceed:", proceed);
    }
}
```

Screenshot of the output
Explanation

4. Final Practical Task: The POS Script

Scenario: Construct a "Point-of-Sale" script to calculate a total bill including a 6% service tax.

1. **Add the HTML UI (The Trigger):**

HTML

```
<section class="lab-card" style="border: 2px solid #27ae60; margin-top:
20px; padding: 15px; border-radius: 8px;">
    <h2 style="color: #27ae60;">Final Task: POS Script</h2>
    <div class="instruction">
        <strong>Scenario:</strong> Calculate a total bill with 6% Service
Tax.
    </div>
    <p>Price per Item: <strong>RM 15.00</strong></p>
    <p>Tax Rate: <strong>6%</strong></p>

    <button onclick="runPOS()" style="background-color: #27ae60; color:
white; padding: 10px 20px; border: none; border-radius: 5px; cursor:
pointer;">
        Checkout Now
    </button>
</section>
```

2. **Add the JavaScript Logic:**

JavaScript

```
/* ==========================================
   FINAL TASK: Point-of-Sale (POS)
   ========================================== */
function runPOS() {
    // Constants
    const pricePerItem = 15;
    const taxRate = 0.06;

    // Input
    let quantity = prompt("How many items do you want to buy?");

    // Validation & Calculation
    if (quantity !== null && quantity !== "") {
        let qtyNumber = parseInt(quantity);

        if (!isNaN(qtyNumber) && qtyNumber > 0) {
            let subtotal = pricePerItem * qtyNumber;
            let totalTax = subtotal * taxRate;
            let finalBill = subtotal + totalTax;

            // Output with 2 decimal places
            alert("Order Summary:\n" +
                  "------------------\n" +
                  "Quantity: " + qtyNumber + "\n" +
                  "Subtotal: RM " + subtotal.toFixed(2) + "\n" +
                  "Service Tax (6%): RM " + totalTax.toFixed(2) + "\n\n" +
                  "Your total bill is: RM " + finalBill.toFixed(2));
        } else {
            alert("Invalid input! Please enter a valid number.");
        }
    }
}
```

What happens when you run this?

1. **Prompt:** Input box pops up. Type 3.
2. **Calculation:** * Subtotal:

   $$3 \times 15 = 45$$
   - Tax:
     $$45 \times 0.06 = 2.70$$
   - Total:
     $$45 + 2.70 = 47.70$$

3. **Output:** An alert box appears: **"Your total bill including tax (6%) is: RM 47.70"**.

Screenshot of the output
Explanation

5. Quick Check Questions

Part A

Question 1

Which of the following correctly describes the difference between the = and + operators in JavaScript?

- A. = is used for addition, while + is used to assign values to variables.
- B. = is the Assignment Operator (stores a value), while + is the Arithmetic or Concatenation Operator.
- C. Both are Arithmetic Operators used for mathematical calculations only.
- D. = compares two values, while + creates a new variable.

Question 2

What is the result of the expression "5" * 2 in JavaScript, and why?

- A. 55, because JavaScript repeats the string.
- B. Error, because you cannot multiply a string with a number.
- C. 10, because JavaScript performs "Type Coercion" to convert the string to a number.
- D. NaN (Not a Number), because the operation is mathematically impossible.

Question 3

In a Point-of-Sale (POS) system, why is it recommended to use const instead of let for a variable like taxRate = 0.06?

- A. const allows the tax rate to be changed easily later in the script.
- B. const makes the variable run faster in the browser memory.
- C. const prevents accidental reassignment, ensuring the fixed tax value remains consistent.
- D. const is the only keyword that supports decimal (float) numbers.

Part B

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1. Which keyword is used to declare a variable whose value is intended to remain constant throughout the program?

A. let

B. var

C. const

D. static

2. What is the output of the following code? console.log(typeof "2026");

A. number

B. string

C. boolean

D. undefined

3. Which operator is used to find the remainder of a division between two numbers?

A. /

B. *

C. &

D. %

4. How do you display a message box to the user that requires them to click "OK" or "Cancel"?

A. alert()

B. prompt()

C. confirm()

D. console.log()

5. Which of the following is a valid Boolean value in JavaScript?

A. "true"

B. Yes

C. false

D. 1

6. What will be the result of the following operation: let x = "5" + 2;?

A. 7

B. 52

C. NaN

D. Error

7. Which method is used to convert a string into a whole number (integer)?

A. parseFloat()

B. toString()

C. parseInt()

D. toNumber()

8. In JavaScript, which operator is used for "Strict Equality" (checking both value and data type)?

A. =

B. ==

C. ===

D. !==

9. What is the purpose of the prompt() function?

A. To display a warning message.

B. To print data to the browser console.

C. To allow the user to input text data.

D. To permanently delete a variable.

10. If let a = 10; and let b = 3;, what is the result of a ** b (Exponentiation)?

A. 13

B. 30

C. 100

D. 1000

JavaScript Quick Tips for Students

Step 1: Variables (`const` vs `let`)

- **const (Constant):** Gunakan untuk nilai yang tidak akan berubah. Contohnya: Nama kursus atau Kadar Cukai. Jika anda cuba ubah nilainya, JavaScript akan bagi error.
- **let:** Gunakan untuk nilai yang boleh berubah-ubah. Contohnya: studentCount (bilangan pelajar boleh bertambah atau berkurang).
- **Boolean:** Hanya ada dua nilai: true atau false. Sangat berguna untuk buat keputusan (Logic).

Step 2: Arithmetic Operators

- **+, -, *, /:** Berfungsi seperti kalkulator biasa.
- **% (Modulus):** Ini bukan peratus! Ia mencari **baki** pembahagian.
  - Contoh: 10 % 3 = 1 (kerana $3 \times 3 = 9$, bakinya adalah 1).
  - Kegunaan: Selalu digunakan untuk semak nombor genap/ganjil.

Step 3: The `typeof` Operator

- JavaScript adalah bahasa *dynamically typed*. Maksudnya, kita tak perlu beritahu komputer jika pemboleh ubah itu teks atau nombor, JavaScript akan "teka" sendiri.
- Gunakan typeof untuk **debug** kod anda jika pengiraan matematik tidak menjadi (mungkin anda tertambah Teks dengan Nombor!).

Step 4: Interaction (`prompt`, `alert`, `confirm`)

- **prompt()**: Cara paling mudah untuk dapatkan input dari pengguna. **Penting:** Input dari prompt sentiasa dalam bentuk **String (Teks)**.
- **alert()**: Untuk beri informasi sehala kepada pengguna.
- **confirm()**: Menghasilkan nilai true (jika tekan OK) atau false (jika tekan Cancel).

Final Step: POS Logic (The "Pro" Tips)

- **parseInt()**: Oleh kerana prompt() memberikan teks, kita perlu "paksa" ia menjadi nombor bulat menggunakan parseInt() supaya kita boleh buat pengiraan matematik.
- **isNaN() (Is Not a Number):** Fungsi keselamatan. Ia menyemak sama ada input pengguna itu nombor sah atau bukan (contoh: jika user taip "ABC", isNaN akan jadi true).
- **.toFixed(2)**: Sangat penting untuk aplikasi kewangan. Ia memastikan nombor kita sentiasa ada 2 tempat perpuluhan (RM 10.00, bukan RM 10).

Lab 3.1.1: JavaScript Fundamentals Quiz

1. Which keyword should you use to declare a variable that will NOT change throughout the script (e.g., a Tax Rate)?

A) `let`

B) `var`

C) `const`

D) `fixed`

2. What is the result of the expression `10 % 4` in JavaScript?

A) 2.5

B) 2

C) 4

D) 0

3. If you use `let input = prompt("Enter a number")` and the user types `5`, what is the data type of `input`?

A) Number

B) Boolean

C) Null

D) String

4. What is the purpose of the `parseInt()` function in the POS script?

A) To round a number to the nearest decimal.

B) To convert a string (text) into an integer (number).

C) To calculate the 6% service tax.

D) To display a message on the screen.

5. Which method is used to format a number to exactly two decimal places (e.g., converting 15 to 15.00)?

A) `toTwo()`

B) `math.round(2)`

C) `toFixed(2)`

D) `formatCurrency()`

✅ Answer Key:

1. **C** (const is for constants)
2. **B** (10 divided by 4 is 2 with a **remainder of 2**)
3. **D** (All prompt inputs are returned as **strings** by default)
4. **B** (It ensures the "text" from the prompt can be used for math)
5. **C** (toFixed(2) handles the decimal formatting)